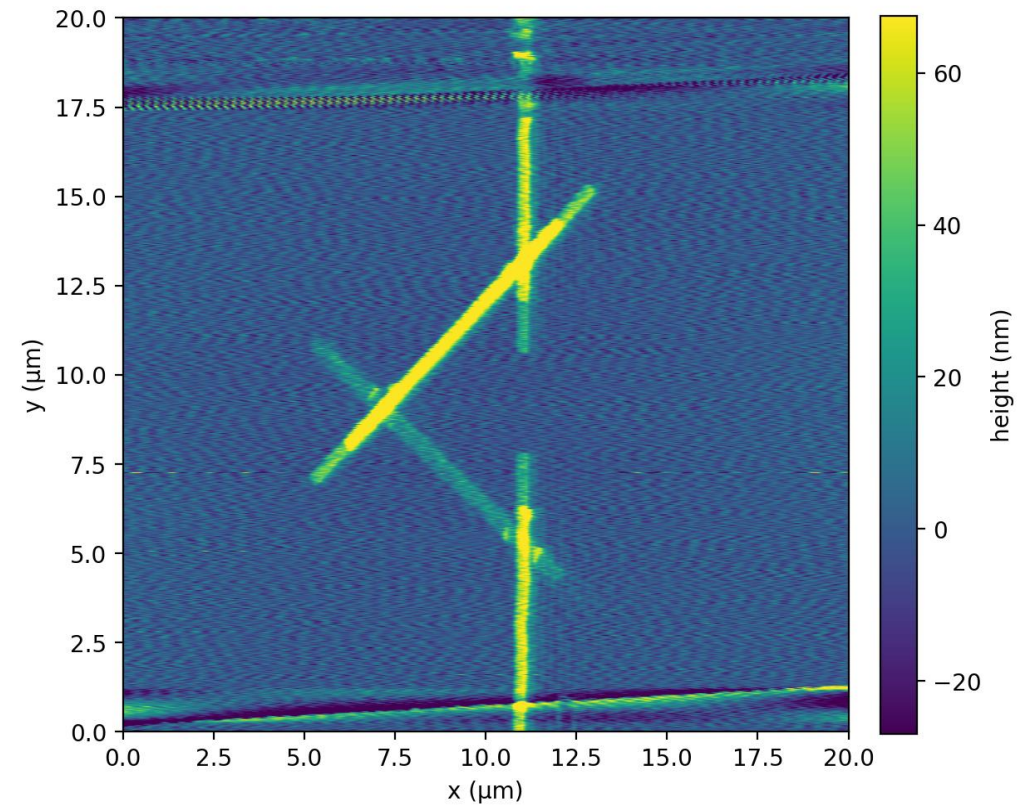
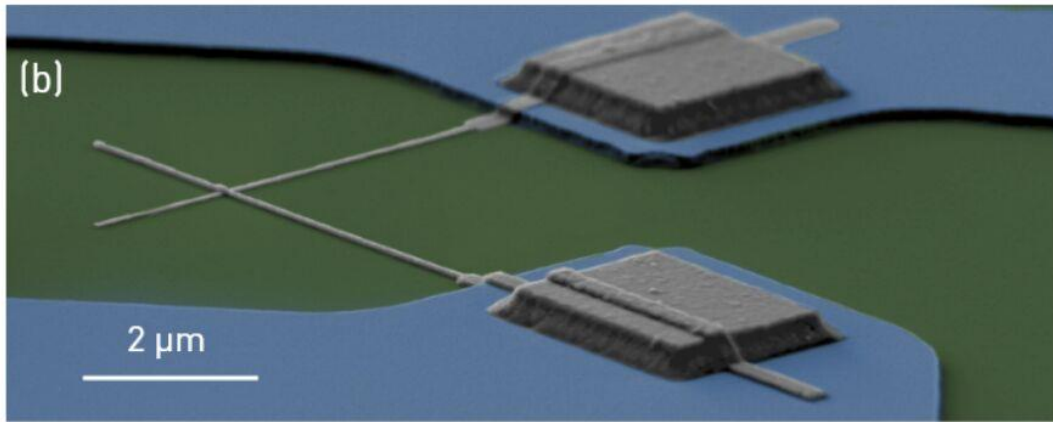
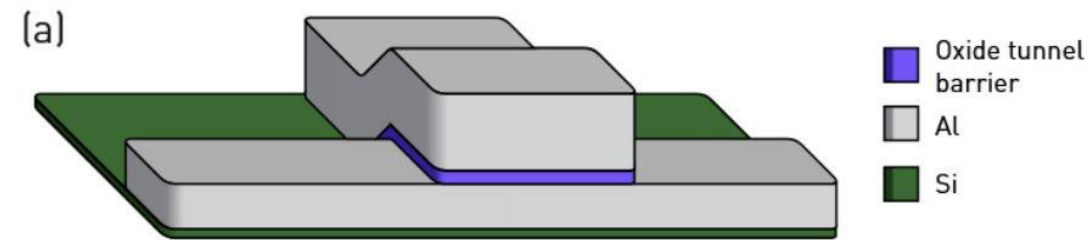
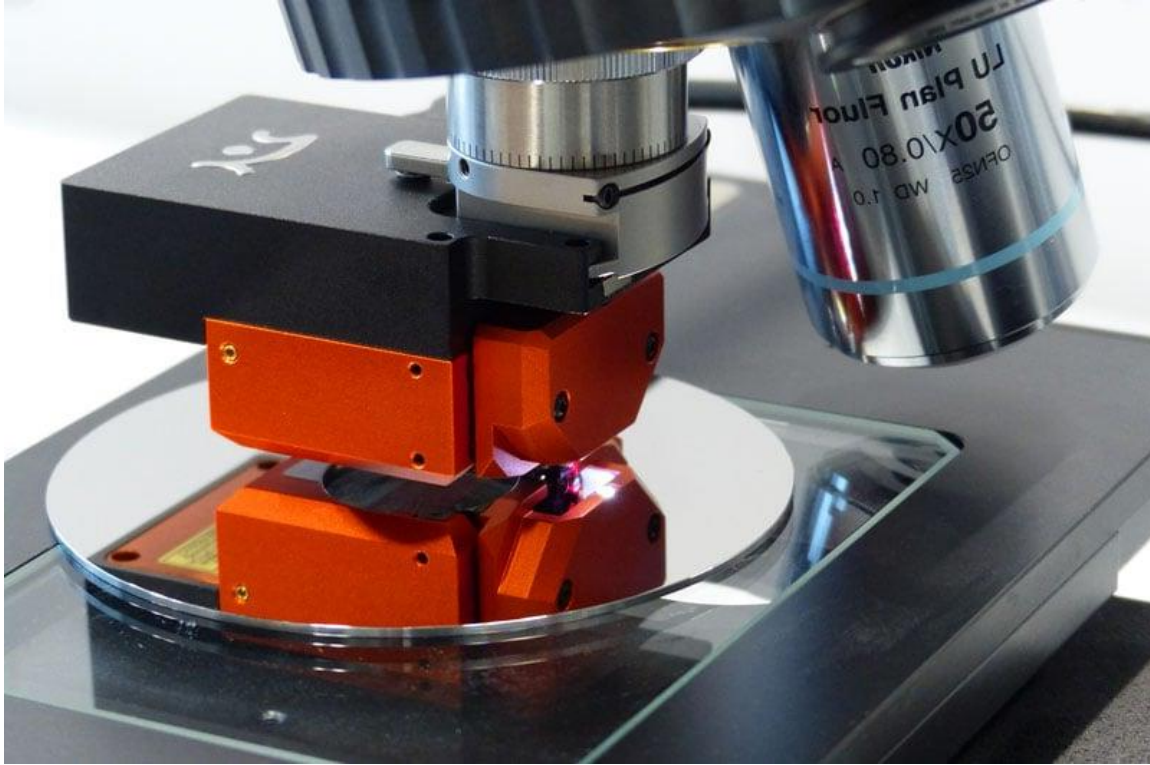




Measurement of Josephson Junction Dimensions using LensAFM

고급물리실험 1조 김민우, 김희민, 오석훈, 최치훈

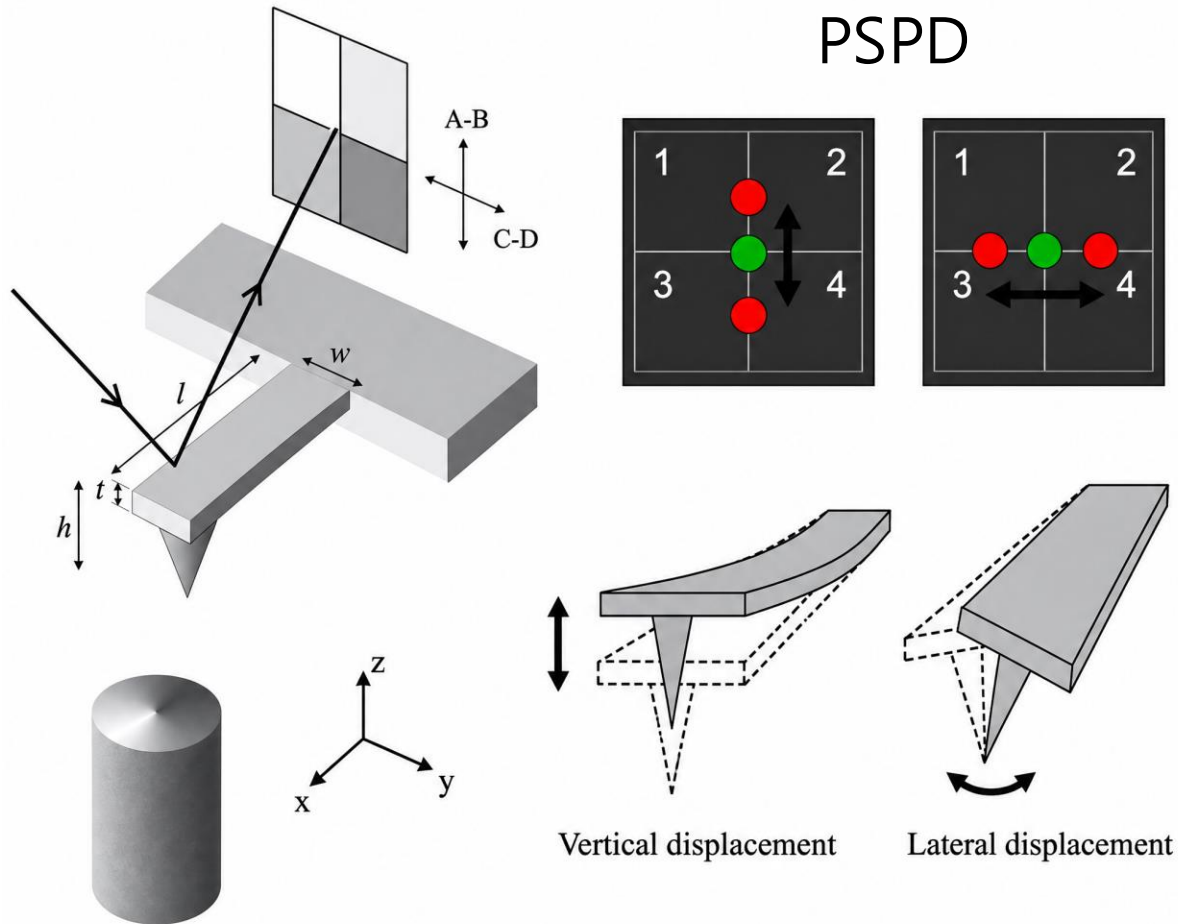
Atomic Force Microscopy (AFM)



Atomic Force Microscopy

- Nanoscale surface imaging technique.
- Provides high-resolution 3D topography.
- Measures roughness and local surface properties.
- Applicable to various materials.
- Minimal sample preparation required.

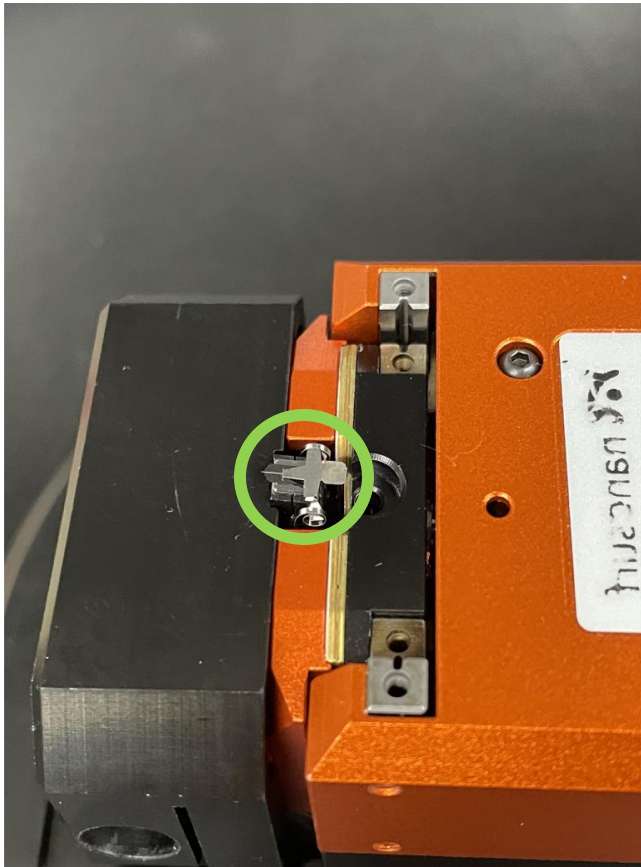
Atomic Force Microscopy (AFM)



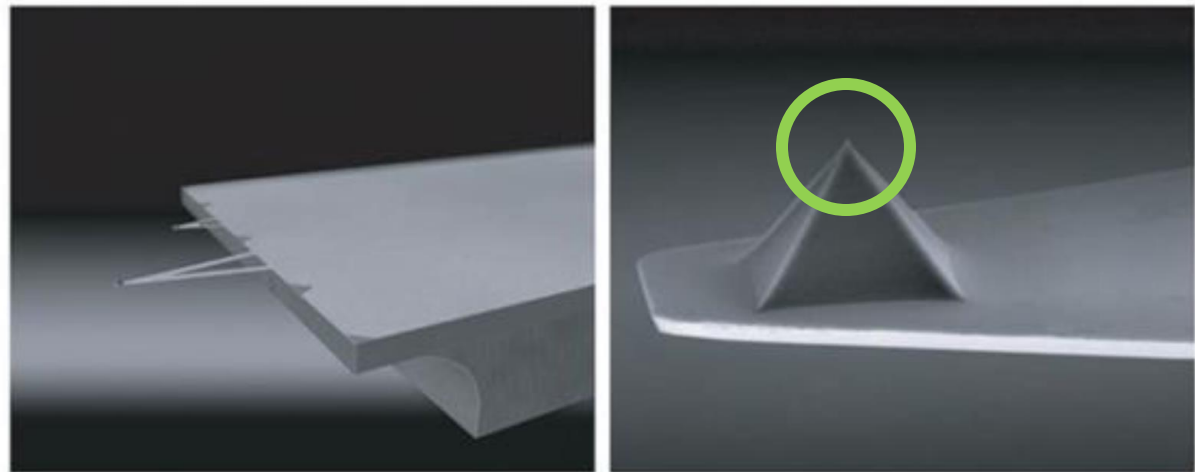
Position-sensitive photodetector (PSPD) can measure displacements of light as small as $10\text{\AA}$

1. A sharp tip scans the sample surface and senses tip-sample forces.
2. These forces bend the cantilever or changes the driven oscillation amplitude.
3. A reflected laser detects the cantilever deflection/amplitude on the PSPD.
4. The signal is used to control the z-piezo to maintain setpoint
5. z-piezo displacement gives topography

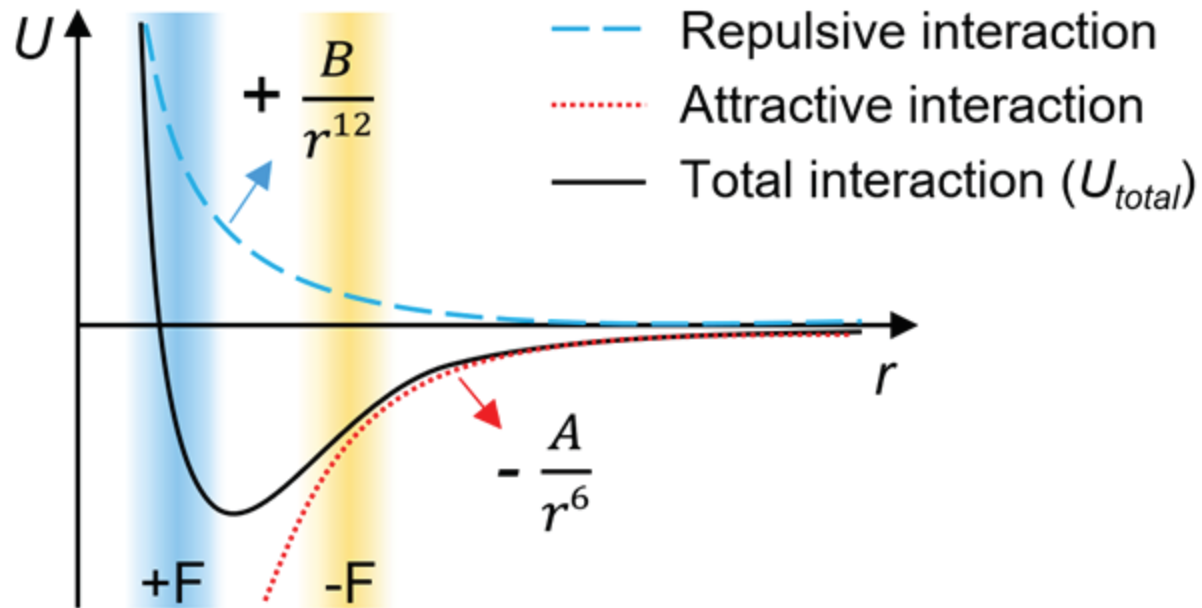
AFM Tip



nm scale Tip edge



AFM Operating mode



- Repulsive regime:
Used in Contact AFM ($F \sim 10^{-3}\text{N} - 10^{-5}\text{N}$)
- Attractive regime:
Used in Non-Contact AFM ($F \sim 10^{-9}\text{N} - 10^{-12}\text{N}$)

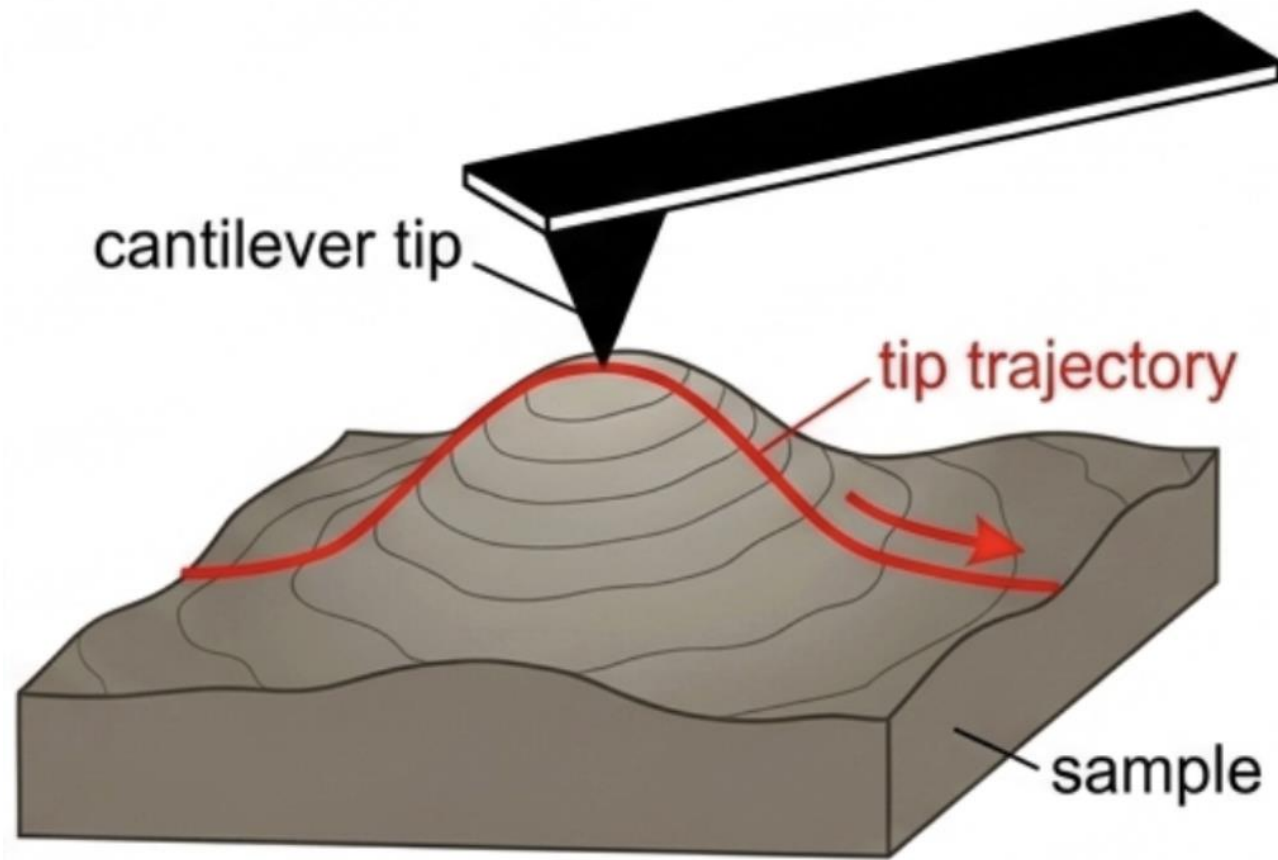
Contact mode

- Very close to the surface (few Å)
- Continuous contact with the sample
- Using repulsive Force

Non-contact mode

- Farther from the surface (~ 100 Å)
- Fixed force gradient
- Using Attractive Force

AFM – Contact mode



Operation : Tip remains in continuous contact with the sample surface.

Measurement: Repulsive tip–sample force bends the cantilever.

Constant-force : Z-scanner maintains constant deflection.

Constant-height: scanner height is fixed; deflection changes are recorded.

Probe Condition: Low spring constant cantilever reduces sample damage.

AFM – Contact mode

Advantages

Fast scanning

High vertical resolution

Suitable for hard and stable surfaces

Limitations

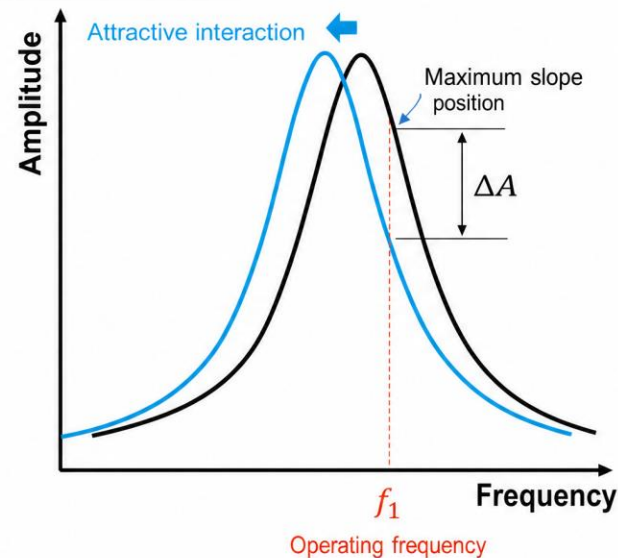
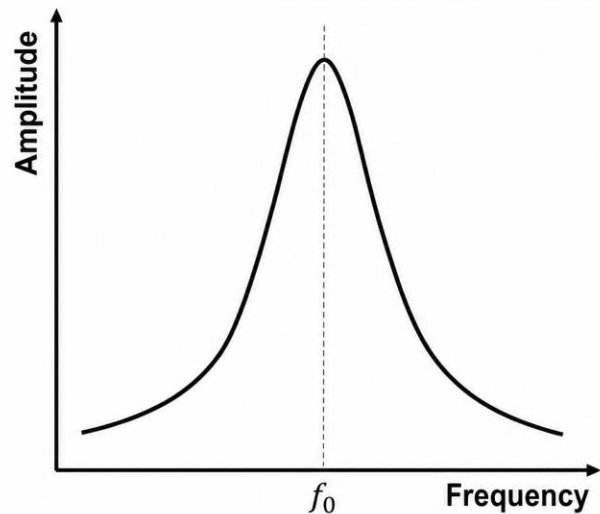
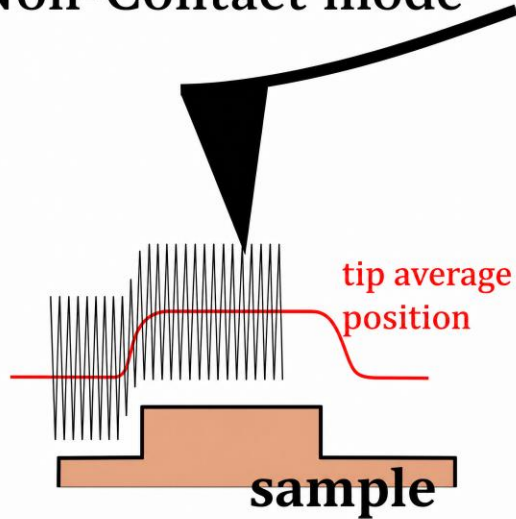
Lateral force can damage soft samples

Tip wear may cause imaging artifacts

Weakly attached objects can be displaced

AFM – Non Contact mode

Non-Contact mode



Tip oscillates above the sample surface near its resonance frequency.

Weak attractive tip–sample forces are detected without direct contact.

As the tip approaches the surface, force gradients shift the resonance frequency. This shift causes changes in the oscillation amplitude or phase.

Z-scanner adjusts tip height to maintain constant tip–sample interaction.

AFM – Non Contact mode

Advantages

Minimal sample damage

Longer tip lifetime

Suitable for soft or delicate samples

Preserves surface condition better

Limitations

Weaker measurement signal

Slower scan speed

Sensitive to vibration and tuning

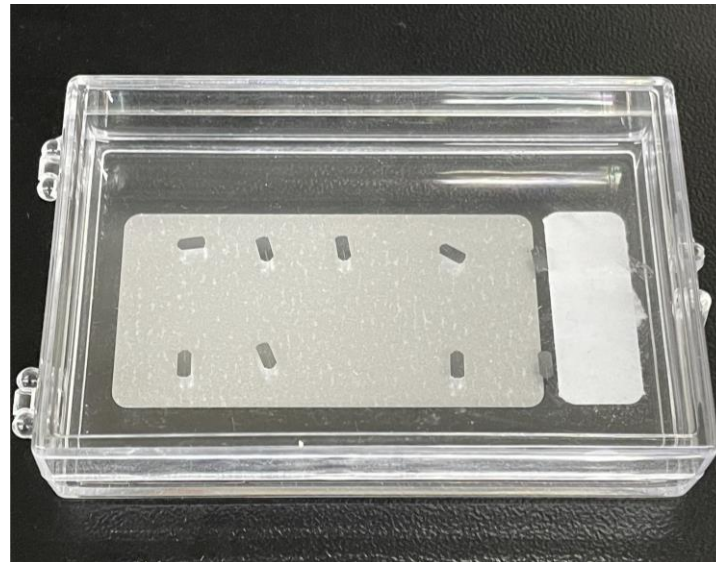
Limited use in liquid due to damping

Methods

AFM preparation



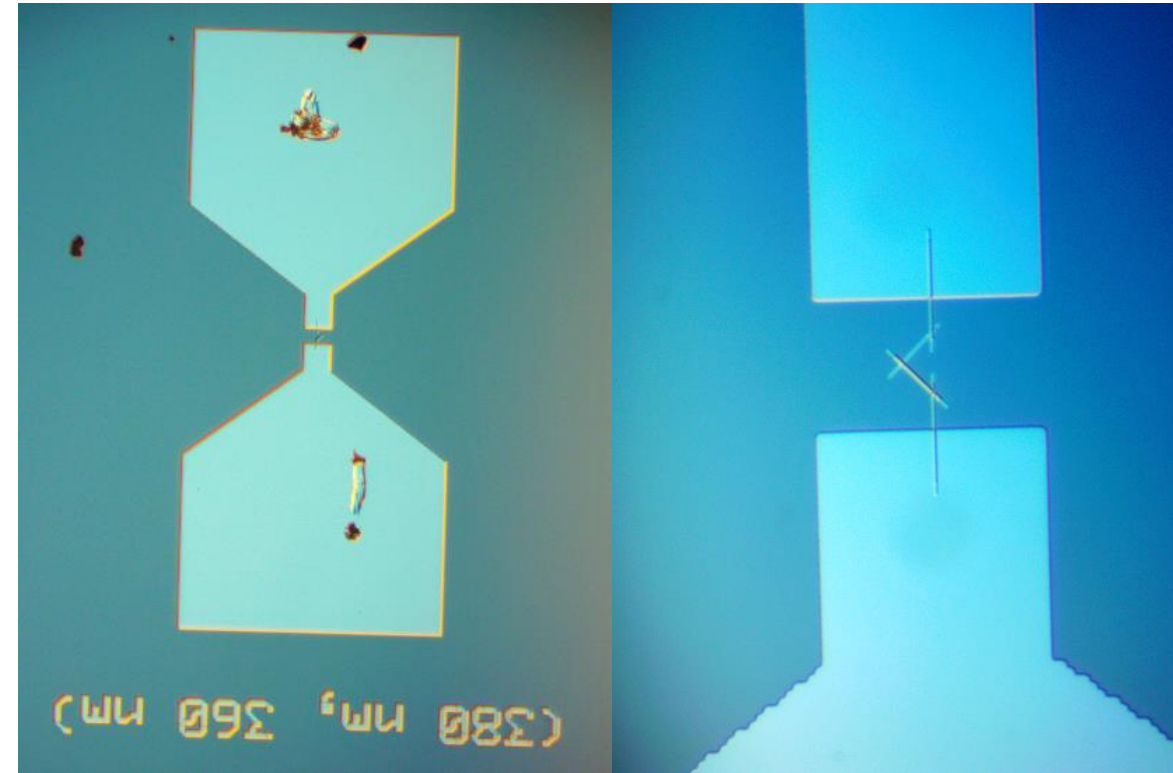
- Check whether the tip of a cantilever is intact
 - Non-contact mode first; since contact mode may damage the sample
- Mount the cantilever onto the AFM using tweezers
- Position the AFM with the objective lens of the optical microscope



Methods

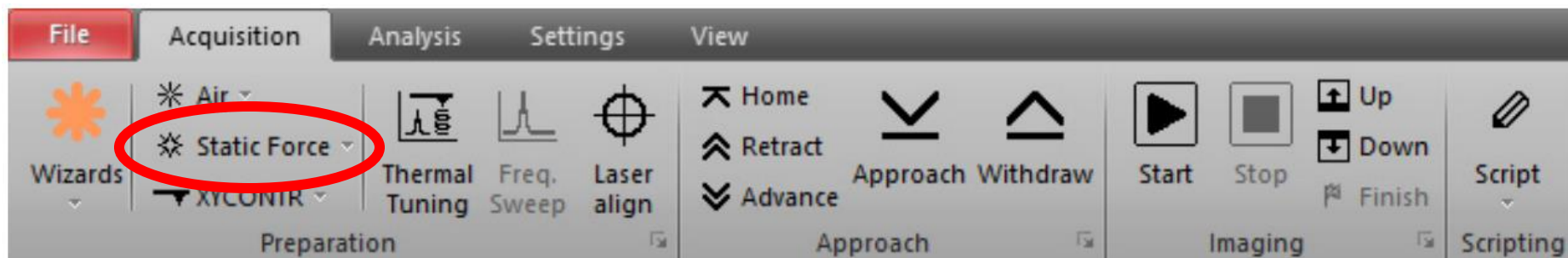
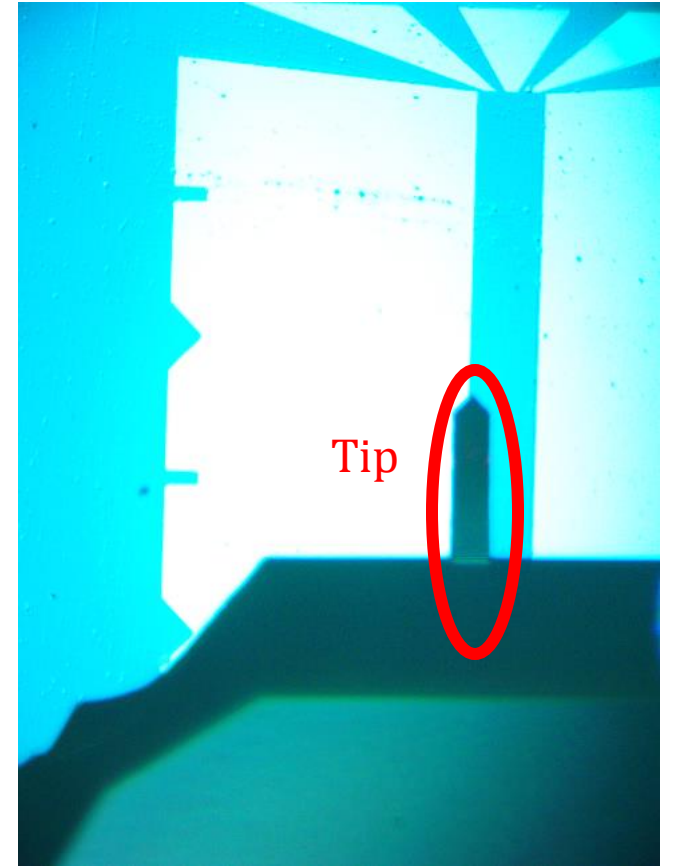
Sample Alignment

- Locate the sample using the optical microscope (20×, 50×, 100×)
- Switch to the AFM objective lens (10×).
- Relocate the target area and adjust the focus.
- Frequency sweep for non-contact mode
→ Determine the cantilever resonance frequency



Methods

- Contact mode ---- Static Force
Non-contact mode ---- Dynamic Force
- Press **Advance** until the tip appears blurry (a double-tip image may be observed)
- Press **Approach**
 - The system automatically positions the tip at the optimal tip–sample distance

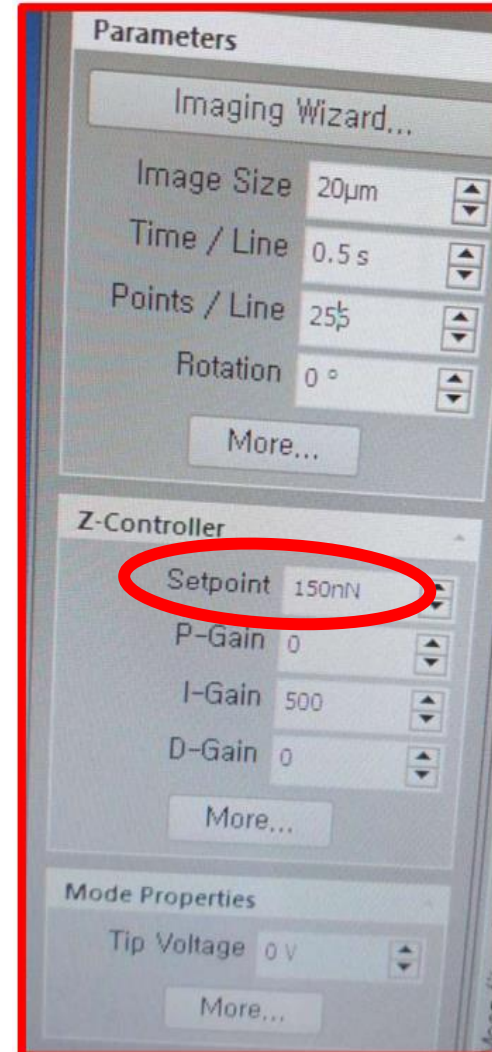


Methods

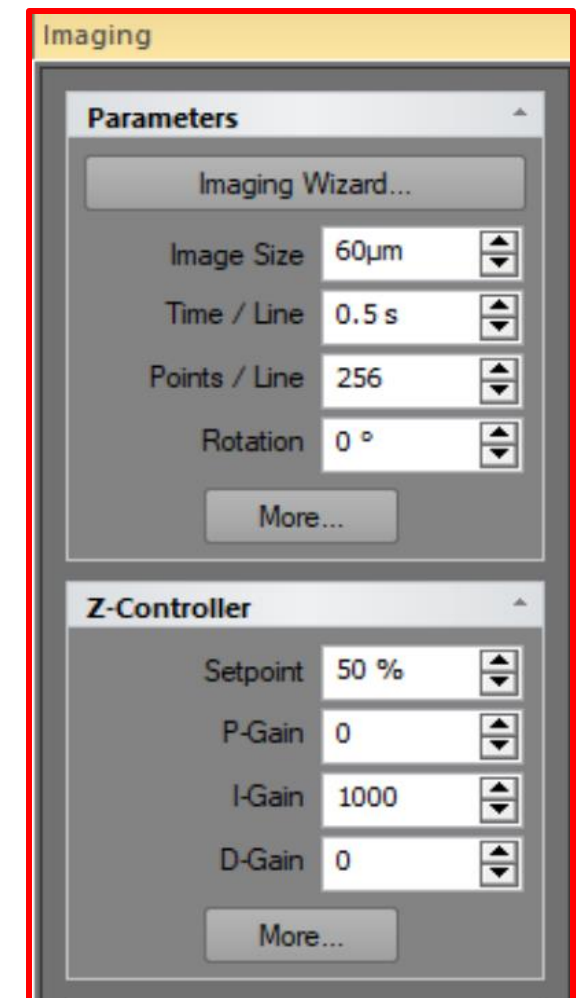
Parameter optimization

- Image Size : Magnification
- Time/Line : Acquisition time(Creep/Drift) vs SNR
- Points/Line : Image resolution
- Setpoint: target cantilever response (Amplitude, deflection) maintained by the feedback loop
- Do not use 150nN setpoint for contact mode for fragile samples

Contact mode

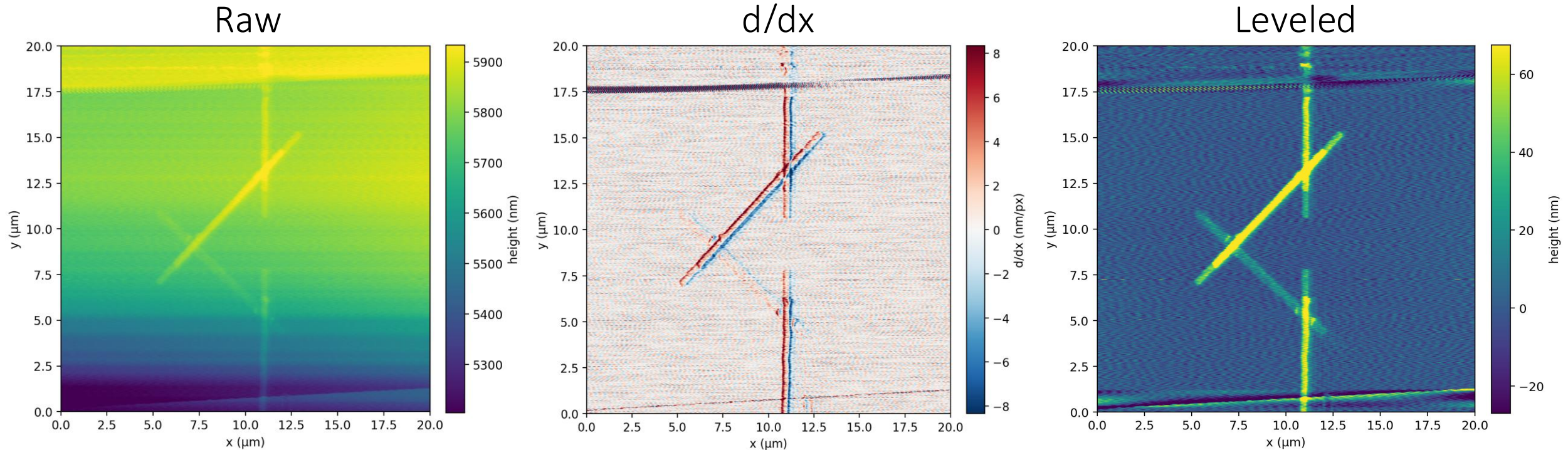


Non-contact mode



Methods: Data Analysis

- Raw z-level from the AFM contains high line to line offset drift (thermal drift etc.)
- Level the image data using per-line masked 2nd polynomial fit removal



Methods: Data Analysis

- Average over a few adjacent lines passing through the electrode to reduce noise
- Error in height estimated from standard deviation of z value of background

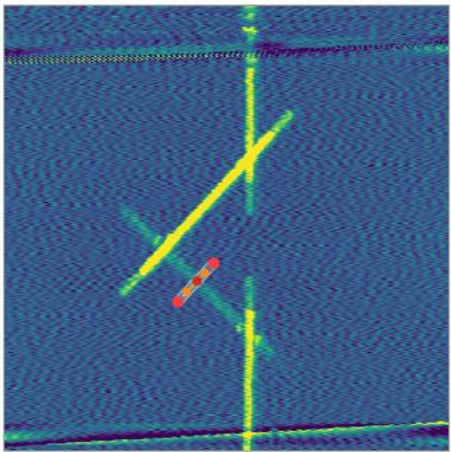
AFM .nid measurement tool — rod height & FWHM (leveled, masked polyfit)

File: Channel: deg: k: right exclude: Image size: Measure on: 1024x1024 px=(19.53, 19.53) nm

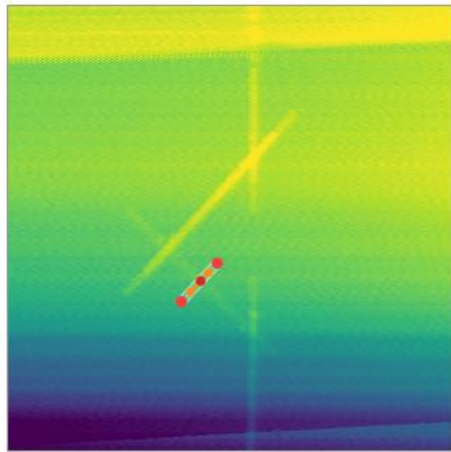
Swath width: Peak average n: FWHM band: sigma ☒ align swath Baseline: Baseline window (each end):

Click two points across a feature on the **leveled** map; drag an endpoint to refine. In **manual** baseline mode, drag the **rod start/end** handles in the profile.

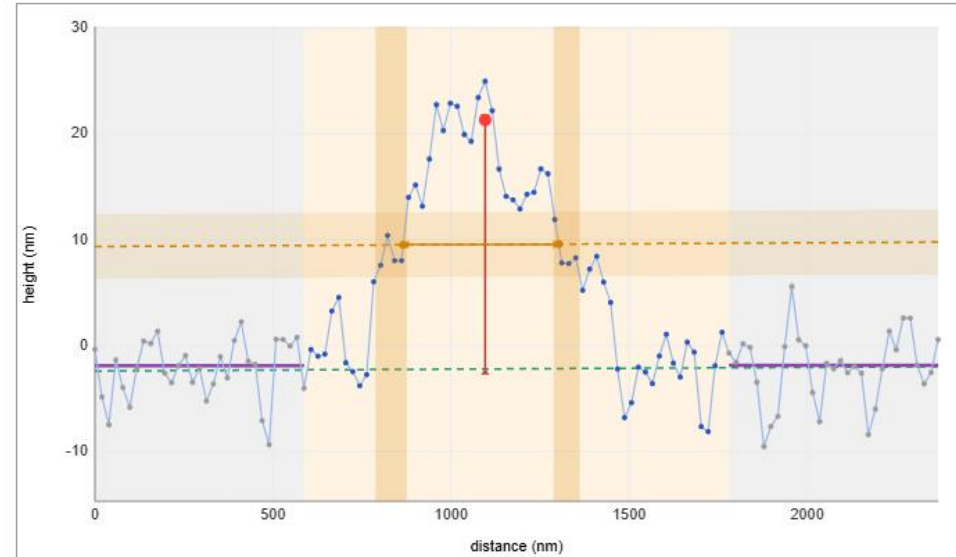
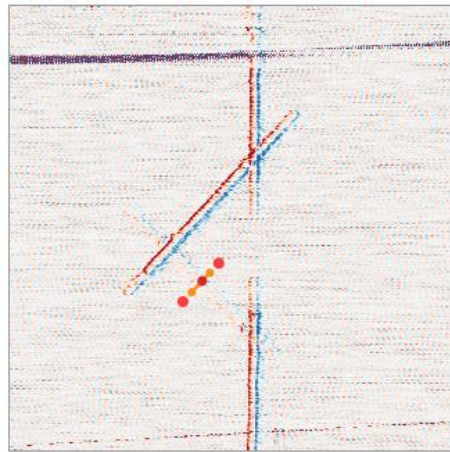
leveled (measure here)



raw (un-leveled)



d/dx (fast scan, raw)



[peak-baseline, peak avg ± 2 pt (5 pts), FWHM band ± 1.00 sigma (3.06 nm), swath 21px aligned, baseline auto]

height **23.51 nm $\pm$ 3.06** | FWHM **436.68 nm $\pm$ 80.69**

Δh (right-left) = -1.85 - -1.89 = **0.03 nm $\pm$ 4.34**

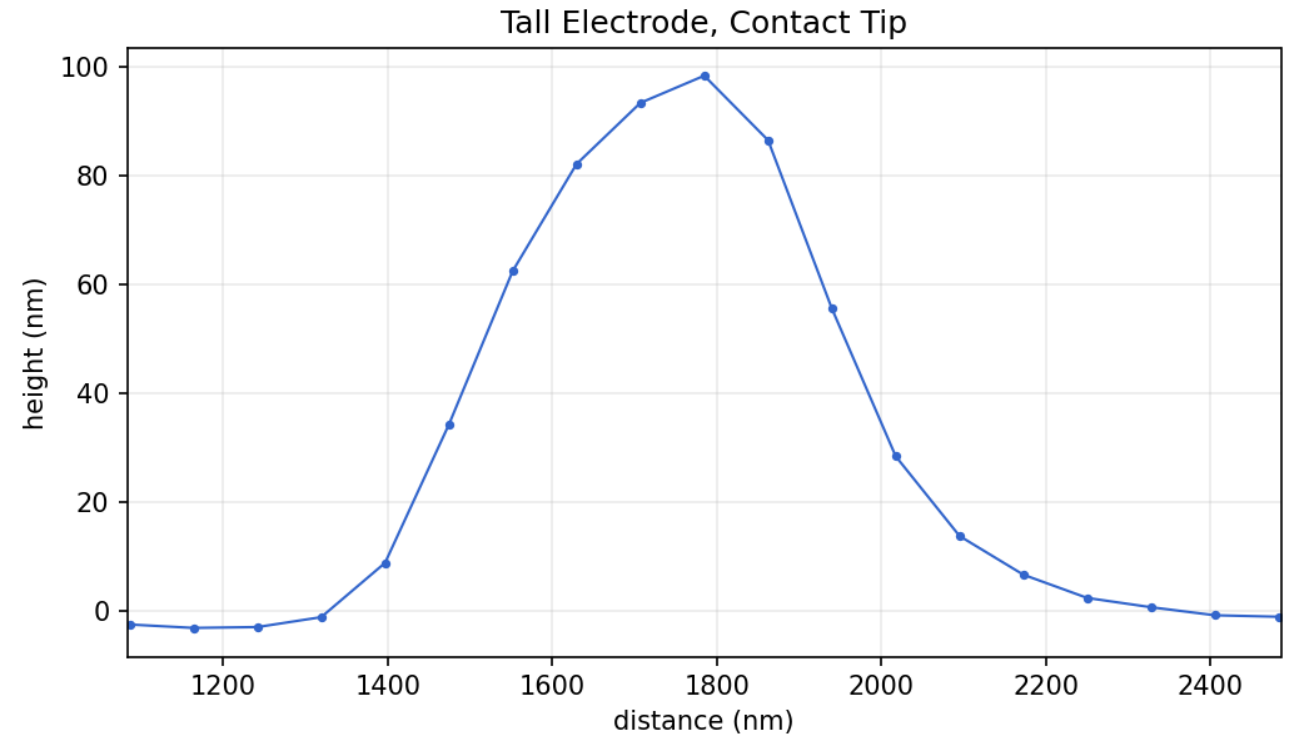
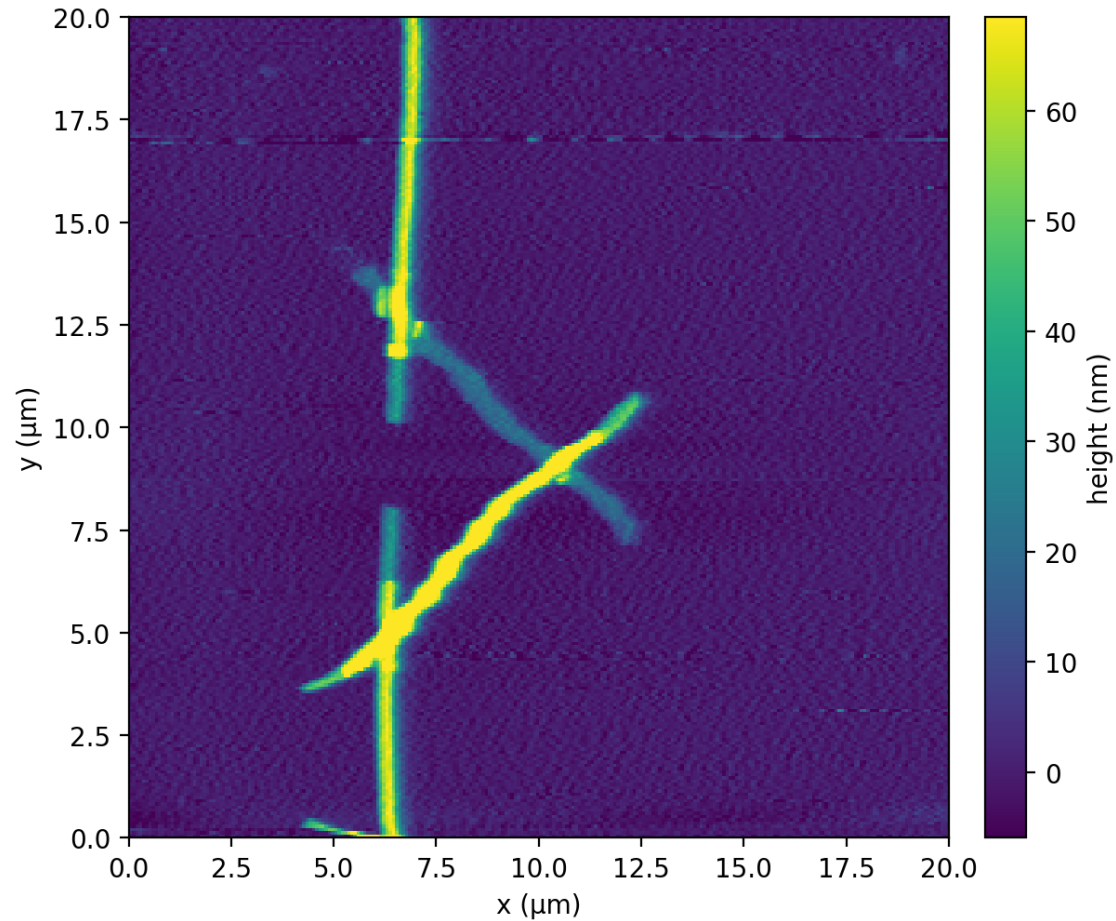
Add measurement ▶

Clear table

Export results

Troubleshooting: Troublesome results

- Plateau is not visible at top of electrodes! Given width of $\sim 500\text{nm}$ scale, tip broadening is insufficient to explain the loss.



Troubleshooting: Troublesome results

- The height doesn't match for different tip measurement

Tip	Electrode	Height (nm)	FWHM (nm)
Contact	Short	21.61 ± 0.40	591.4 ± 9.8
	Tall	95.39 ± 0.52	459.7 ± 2.7
Non-contact	Short	21.48 ± 0.58	726.6 ± 10.3
	Tall	100.19 ± 0.81	819.0 ± 4.8

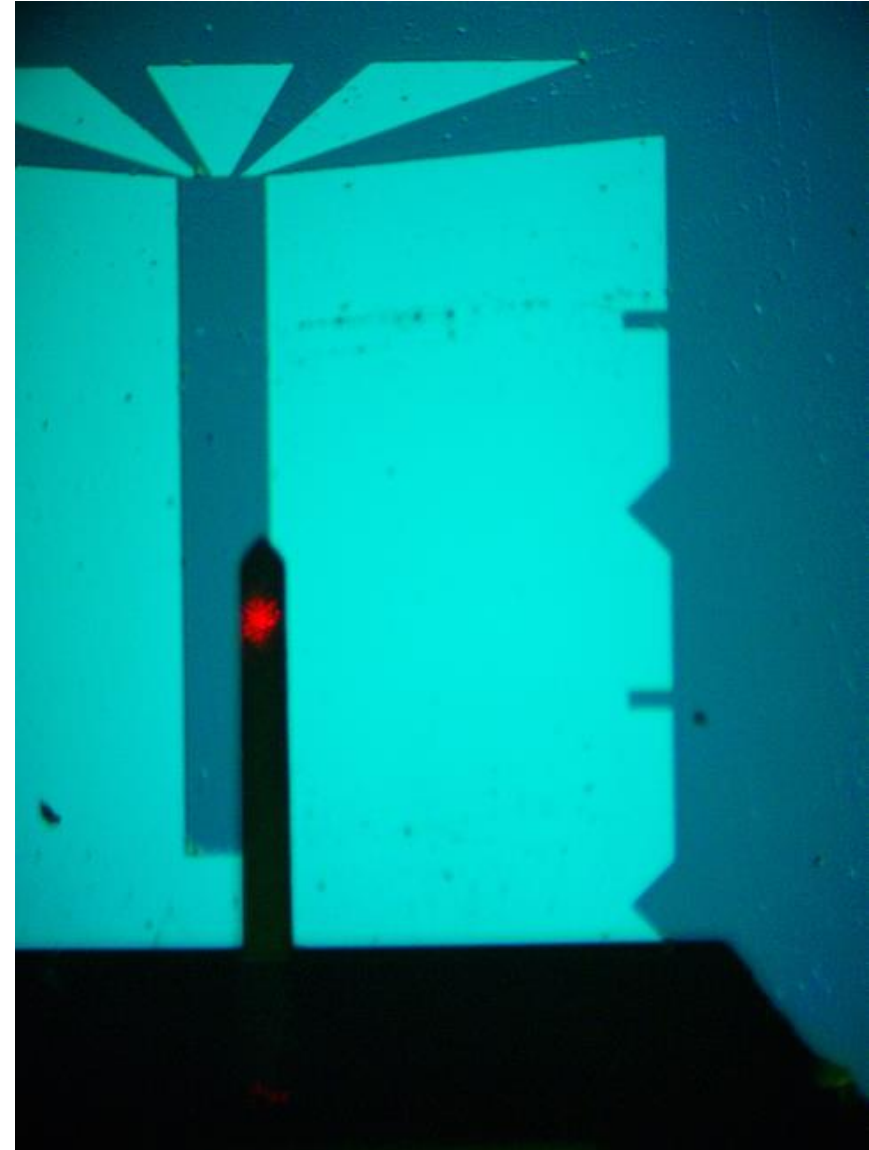
- First run used 256 points for 20μm image, leading to 1 pixel = 78nm (low resolution)
→ Finer imaging (4x finer, 1 pixel = 20nm) by increasing to 1024 data points

Troubleshooting: z-calibration

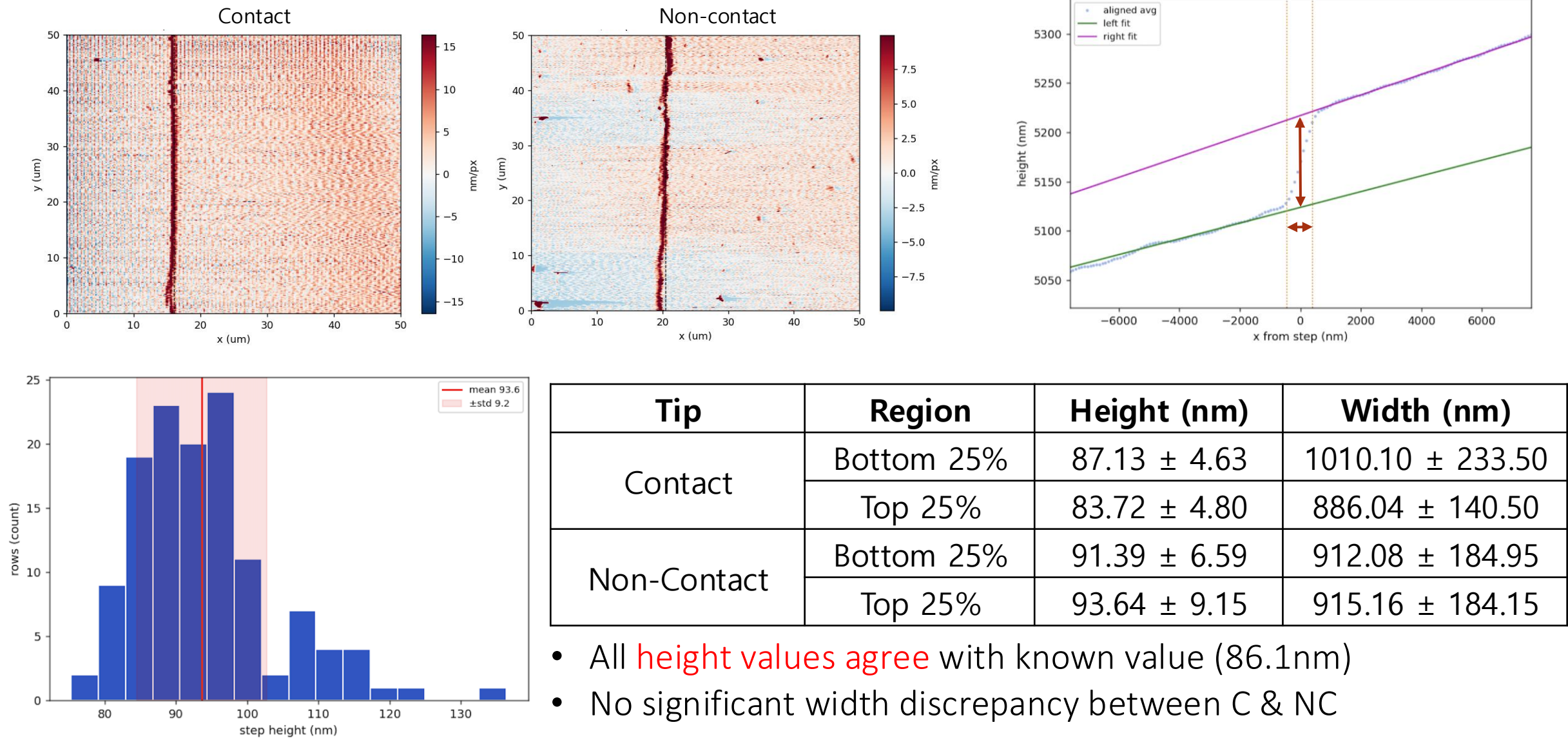
To ensure our height measurements are trustworthy:

- Measure the 'height' of a known sample with higher resolution:

Quartz Step Height Standard - 86.1nm

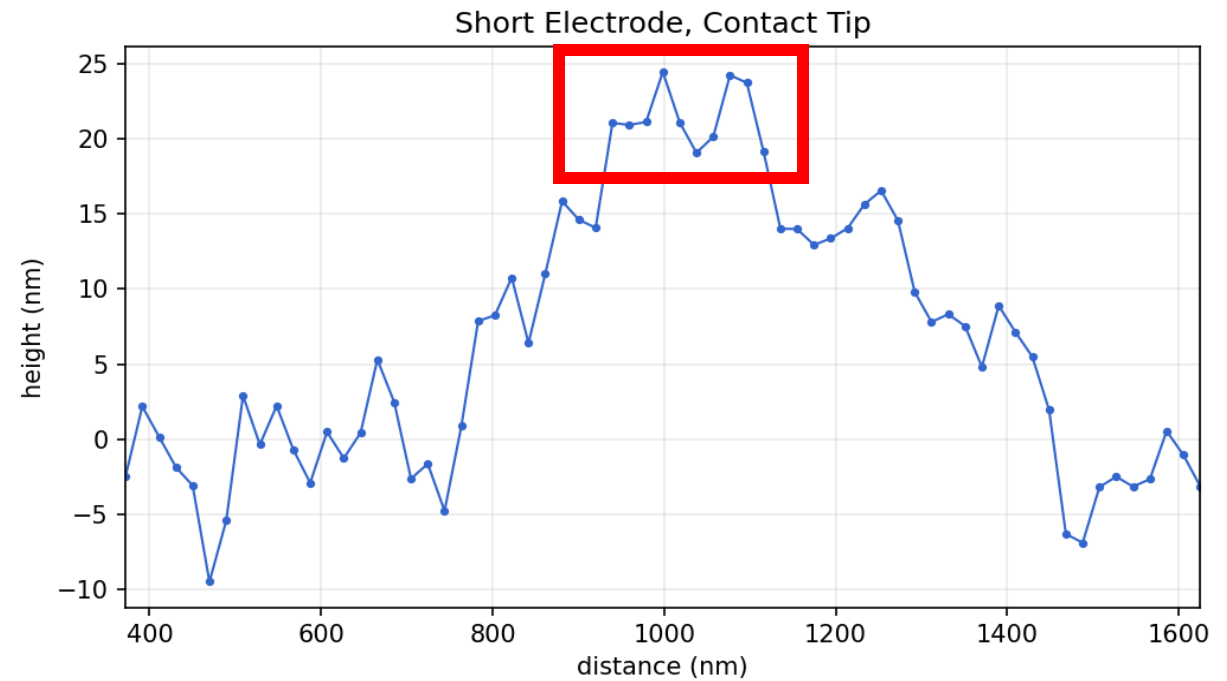
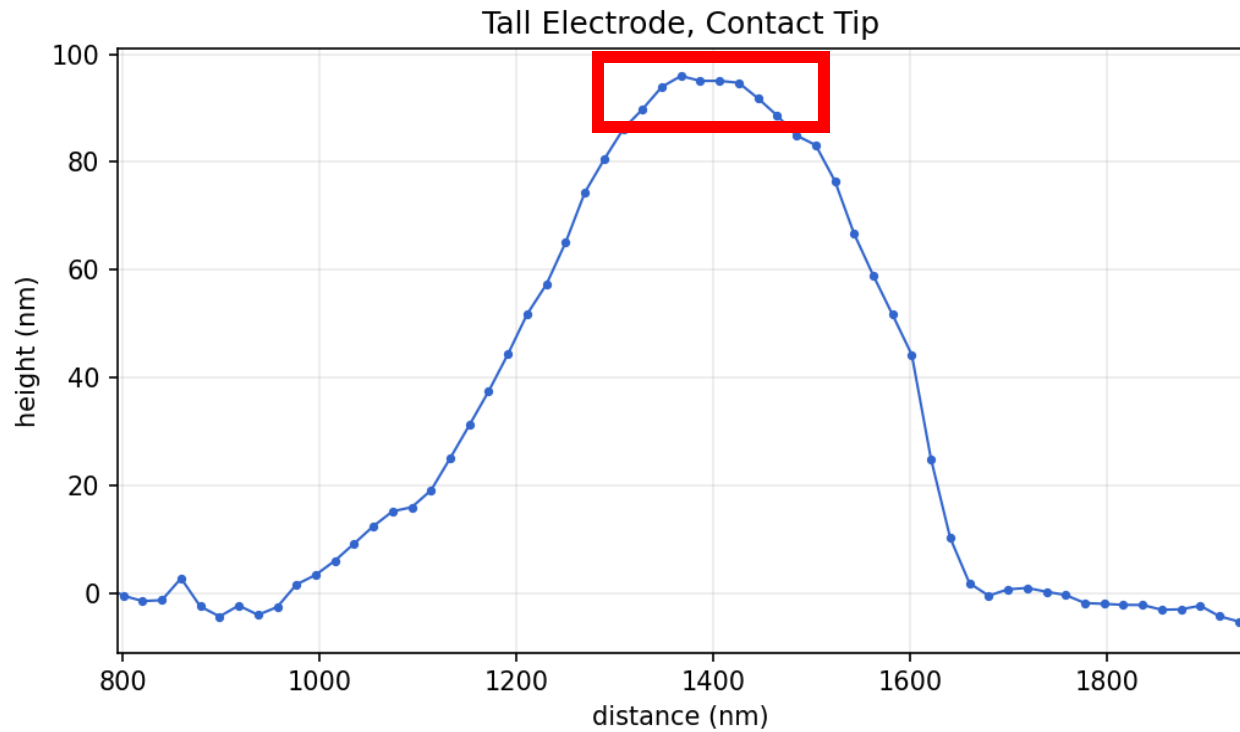


Troubleshooting: z-calibration



Results

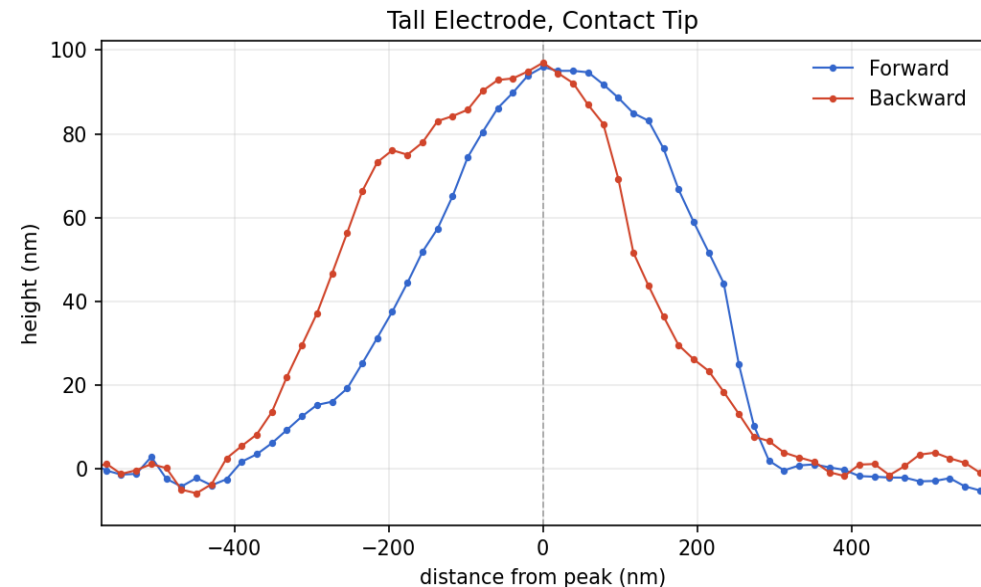
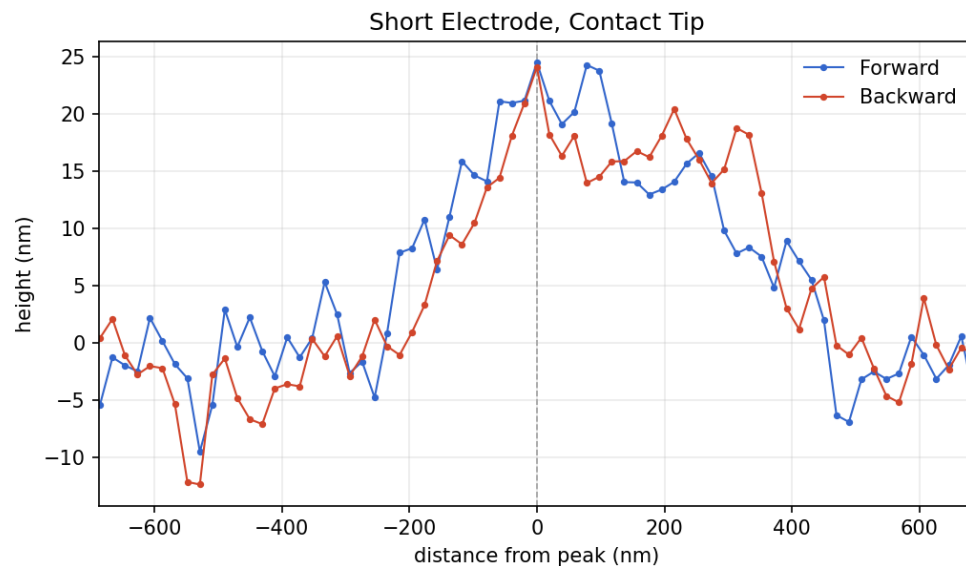
- With finer sampling, plateau visible at the top of the electrodes.



Results

- Forward/backward data gives identical values within experimental error.
- Height matches cleanly but FWHM cleanly differs

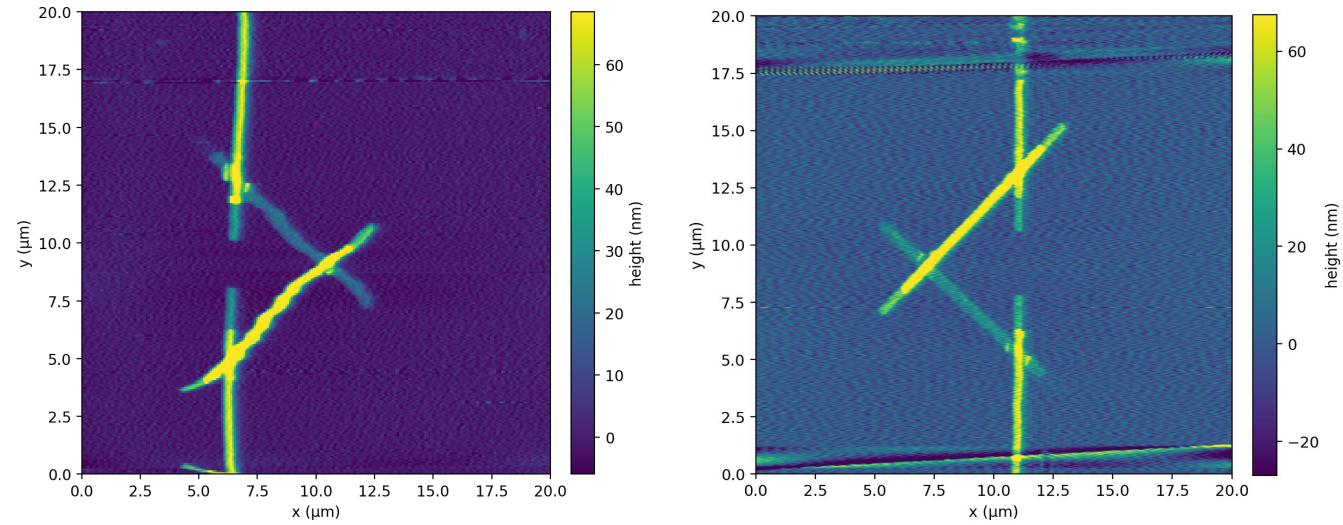
Tip	Electrode	Height (nm)	FWHM (nm)
Contact	Short	23.51 ± 3.06	436.7 ± 80.7
	Tall	95.89 ± 1.89	400.4 ± 9.1
Non-contact	Short	21.76 ± 2.93	765.1 ± 203.8
	Tall	95.20 ± 2.83	929.4 ± 20.8



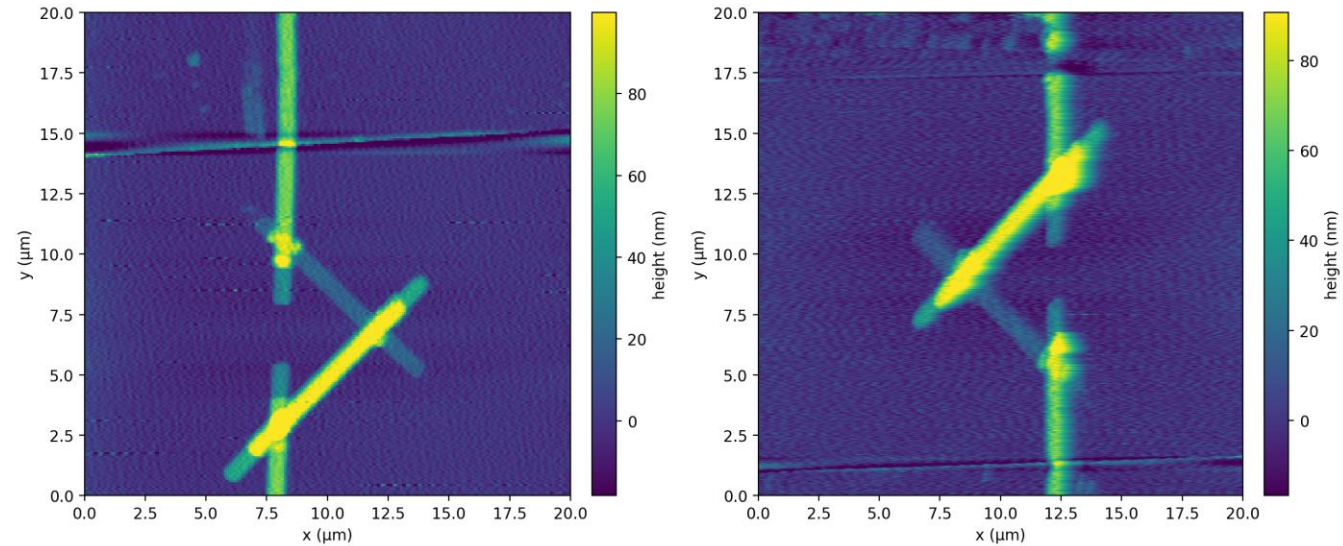
Results

- Even with different tip & resolution, non-contact tip case has higher FWHM.

Contact



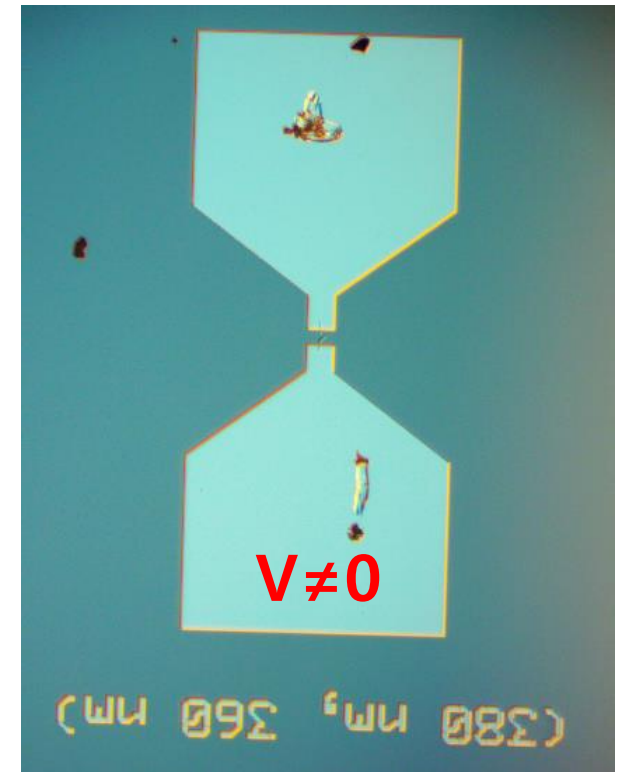
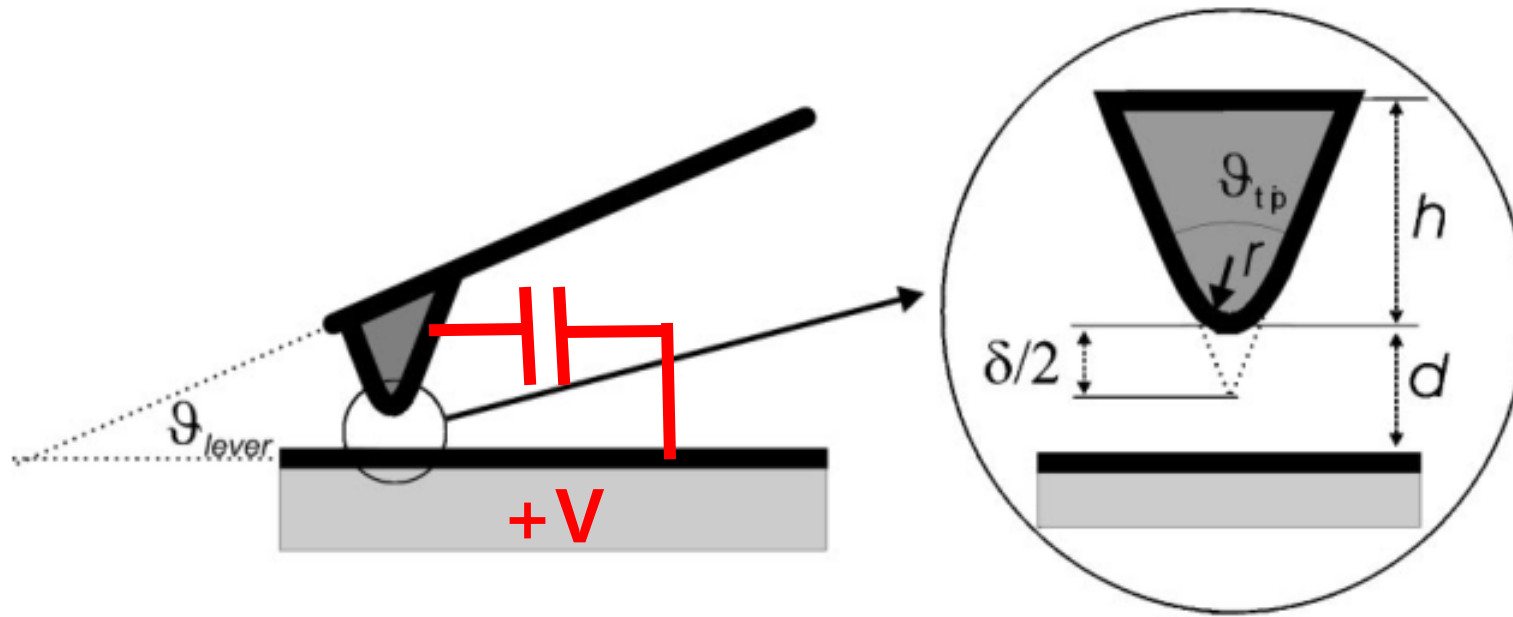
Non-contact



Discussion

What is the origin of width broadening in non-contact mode, only evident in Al electrode sample (not in quartz sample)?

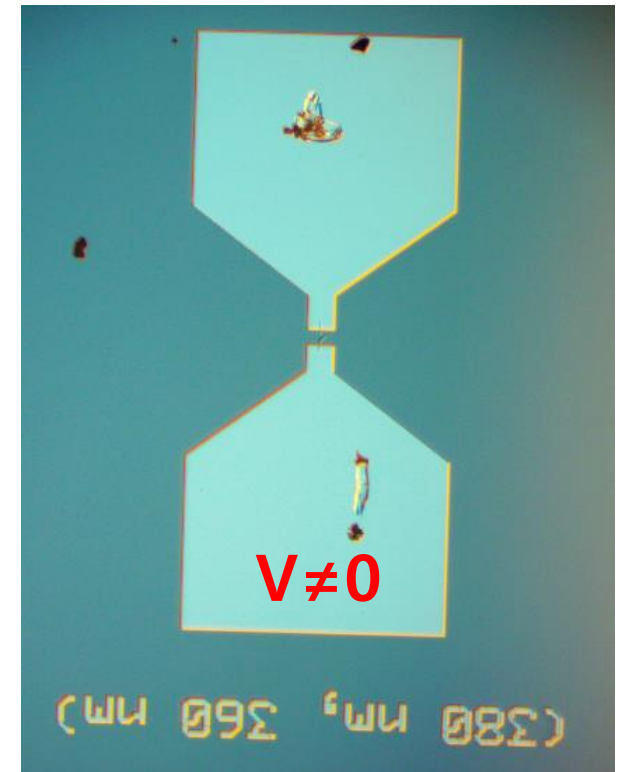
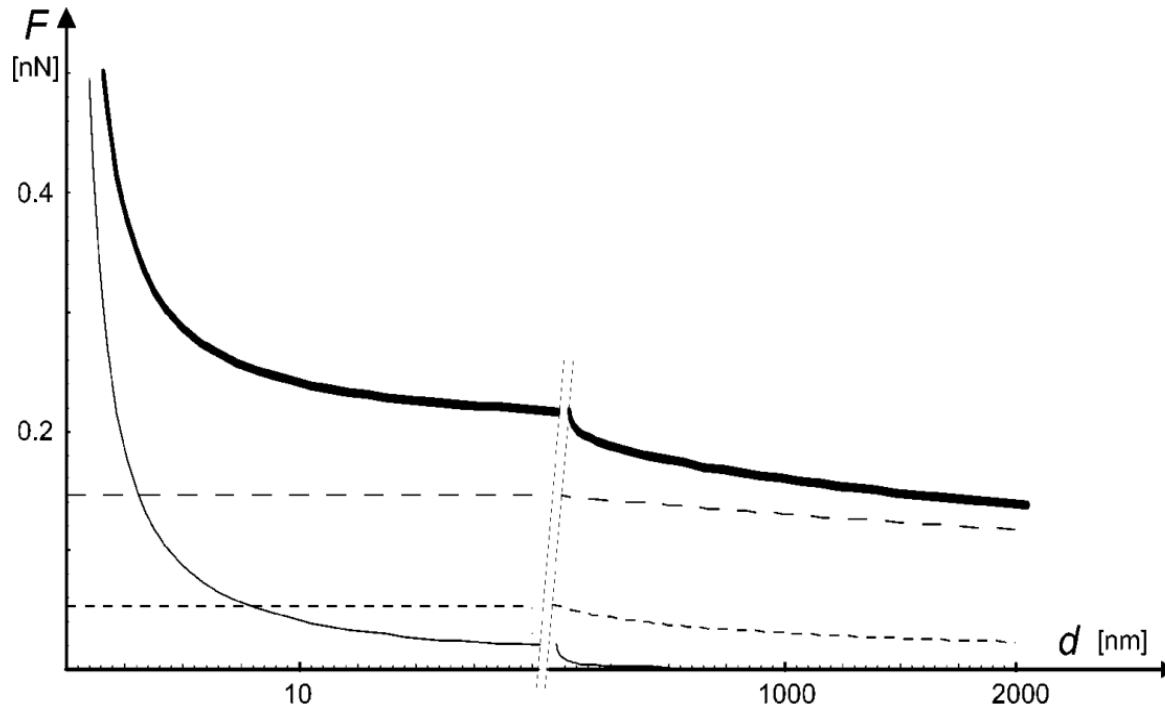
- Electrostatic forces due to ungrounded Al electrode
- Capacitive coupling is quite long ranged; $O(1000\text{nm})$.



Discussion

What is the origin of width broadening in non-contact mode, only evident in Al electrode sample (not in quartz sample)?

- Electrostatic forces due to ungrounded Al electrode
- Capacitive coupling is quite long ranged; $O(1000\text{nm})$.



Discussion

Oscillation is evident!

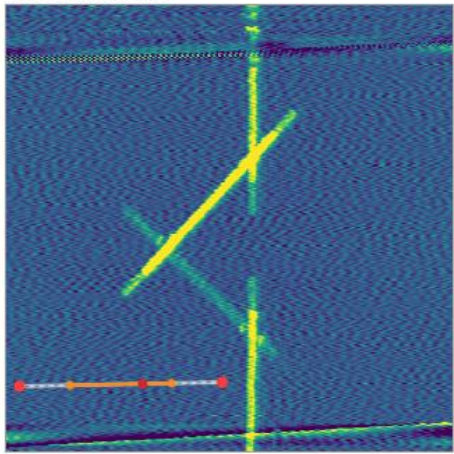
AFM .nid measurement tool — rod height & FWHM (leveled, masked polyfit)

File: Channel: deg: k: right exclude: Image size: Measure on: 1024x1024 px=(19.53, 19.53) nm

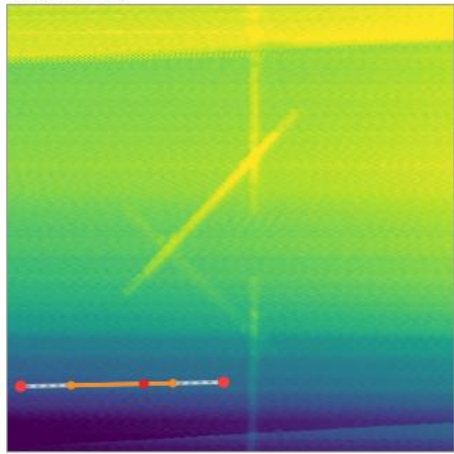
Swath width: Peak average n: FWHM band: sigma ☒ align swath Baseline: Baseline window (each end):

Click two points across a feature on the **leveled** map; drag an endpoint to refine. In **manual** baseline mode, drag the **rod start/end** handles in the profile.

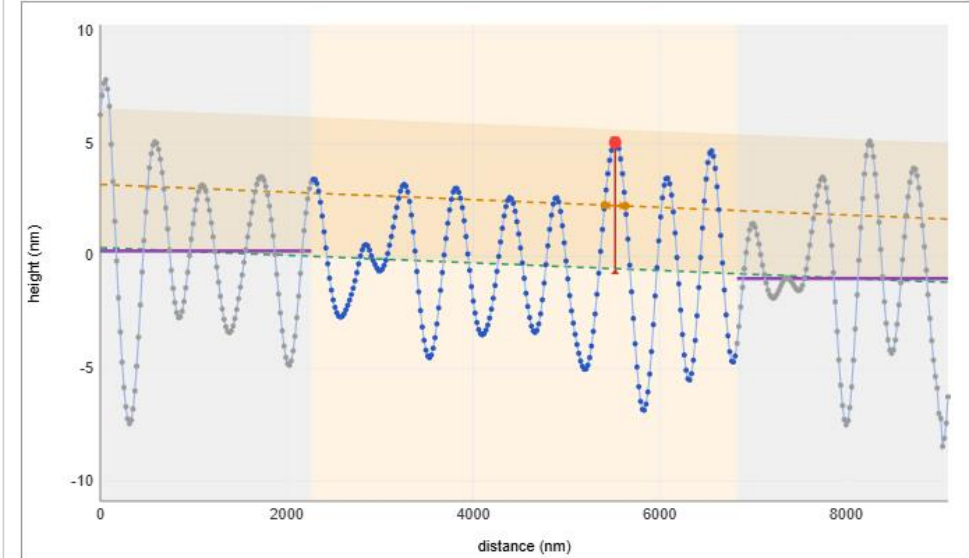
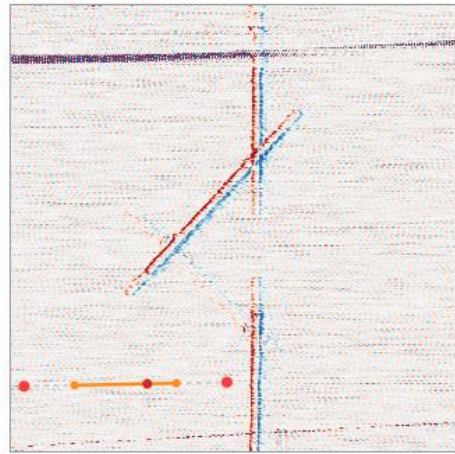
leveled (measure here)



raw (un-leveled)



d/dx (fast scan, raw)



[peak-baseline, peak avg ± 1 pt (3 pts), FWHM band ± 1.00 sigma (3.40 nm), swath 7px aligned, baseline auto]

height **5.62 nm $\pm$ 3.40** | FWHM **222.81 nm $\pm$ NaN**

Δh (right-left) = $-0.99 - 0.24 =$ **-1.23 nm $\pm$ 4.82**

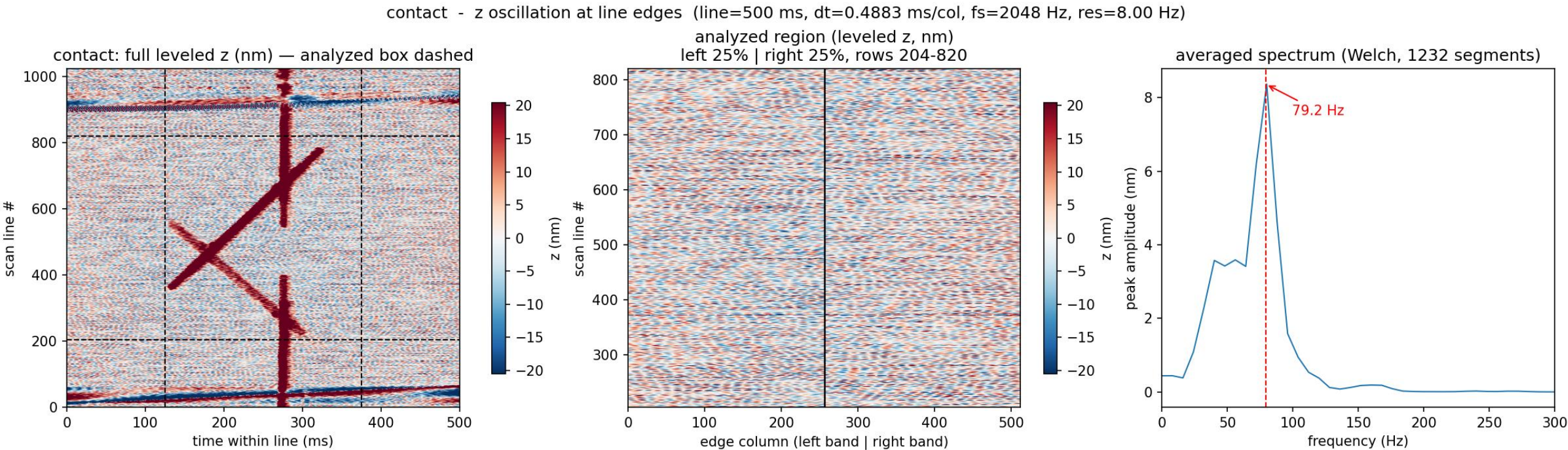
Add measurement ▶

Clear table

Export results

Discussion

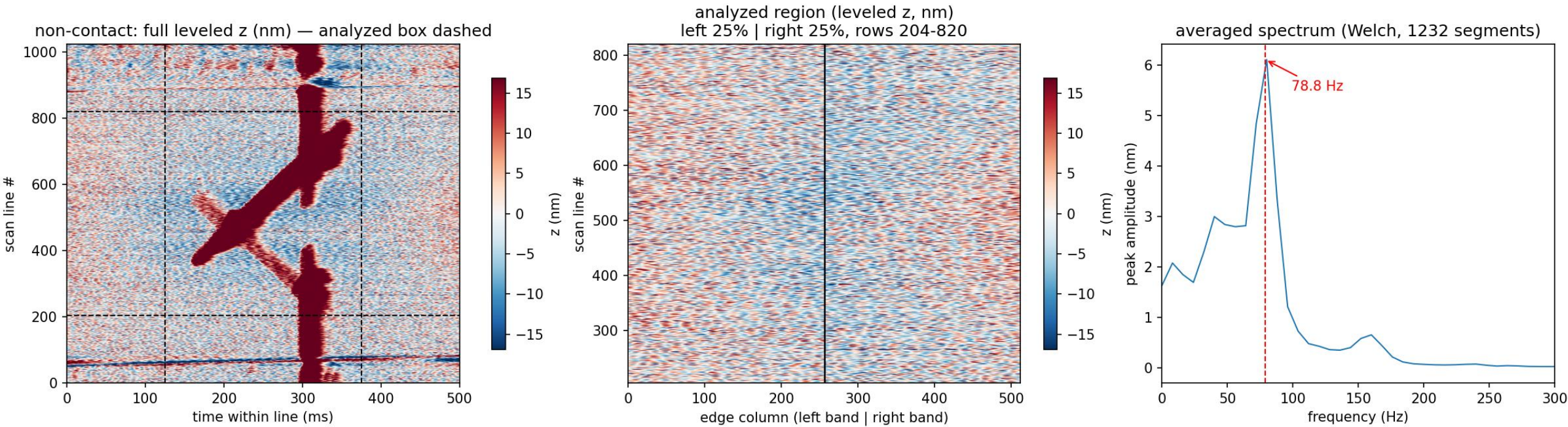
FFT per line, and average the power spectrum → **~80Hz peak**



Discussion

FFT per line, and average the power spectrum → **~80Hz peak**

non-contact - z oscillation at line edges (line=500 ms, dt=0.4883 ms/col, fs=2048 Hz, res=8.00 Hz)



Conclusion

- Height measurements were consistent between contact and non-contact mode within experimental error ($\sim 22\text{nm}$, $\sim 95\text{nm}$).
- Width measurements showed significant difference between the two modes.
- Contact-mode results($\sim 400\text{nm}$, $\sim 440\text{nm}$) were closer to the design value(360nm , 380nm), while the broadening in non-contact mode is likely caused by electrostatic tip–sample interactions.
- AFM is a well-suited tool for height measurement, while lateral dimension measurements can be less accurate.